Low Birth Weight Babies

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Low birth weight (LBW)

- Definition : Birth weight <2500 g

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LBW: Significance

- 75% neonatal deaths and 50% infant deaths occur among LBW infants
- LBW babies are more prone to:
  - Malnutrition
  - Recurrent infections
  - Neuro developmental delay

LBW babies have higher mortality and morbidity

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Types of LBW

2 types based on the origin

Preterm	Small-for-date (SFD) / intra uterine growth retardation (IUGR)
< 37 completed weeks of gestation	< 10th centile for gestational age
Account for 1/3rd of LBW	Account for 2/3rd of LBW neonates

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Causation: LBW

- Etiology of prematurity
  - Low maternal weight, teenage / multiple pregnancy
  - Previous preterm baby, cervical incompetence
  - Antepartum hemorrhage, acute systemic disease
  - Induced premature delivery
  - Majority unknown

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Causation: LBW

Etiology of SFD / IUGR

- Poor nutritional status of mother
- Hypertension, toxemia, anemia
- Multiple pregnancy, post maturity
- Chronic malaria, chronic illness
- Tobacco use

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LBW: Identification of types

Prematurity

- Date of LMP
- Physical features
  - Breast nodule
  - Genitalia
  - Sole creases
  - Ear cartilage / recoil

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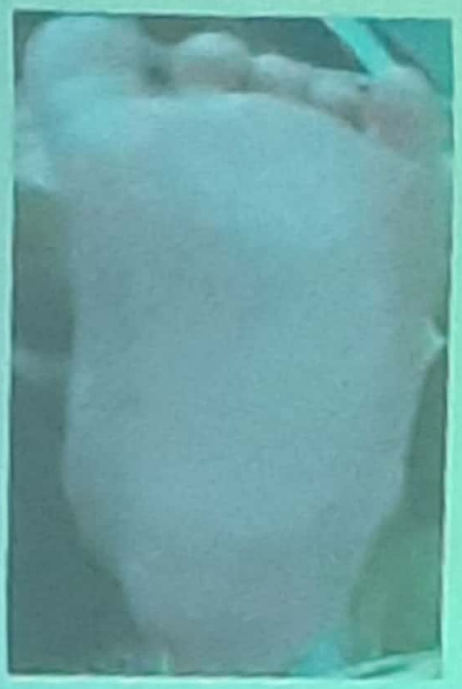
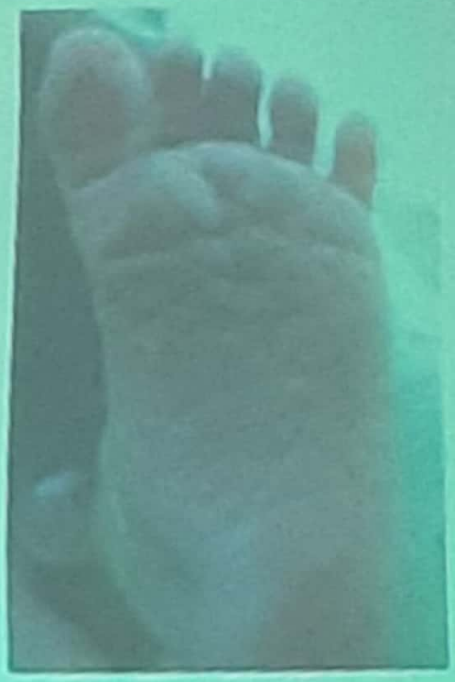
Identification: Preterm LBW

Sole creases

Preterm



Term



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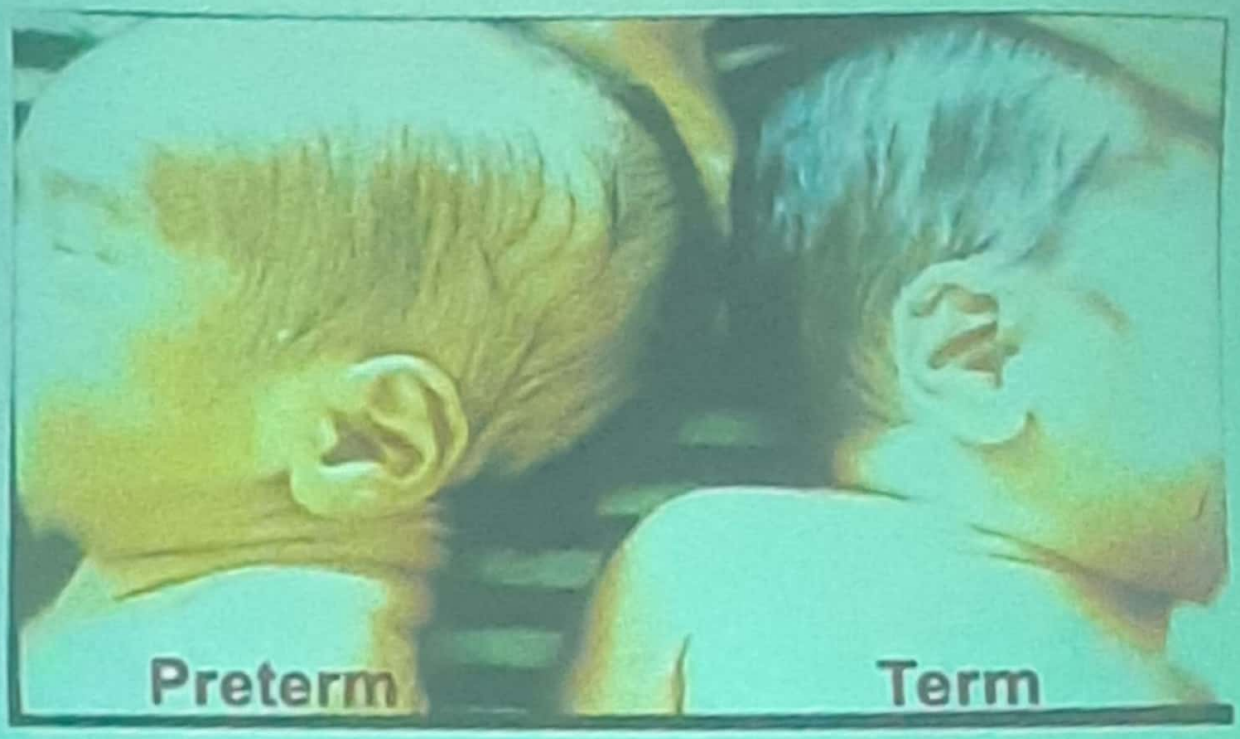
Identification: Preterm LBW

Ear Cartilage



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LBW: Identification of types

SFD / IUGR

- Intrauterine growth chart
- Physical characteristics
  - Emaciated look
  - Loose folds of skin
  - Lack of subcutaneous tissue
  - Head bigger than chest by >3cm

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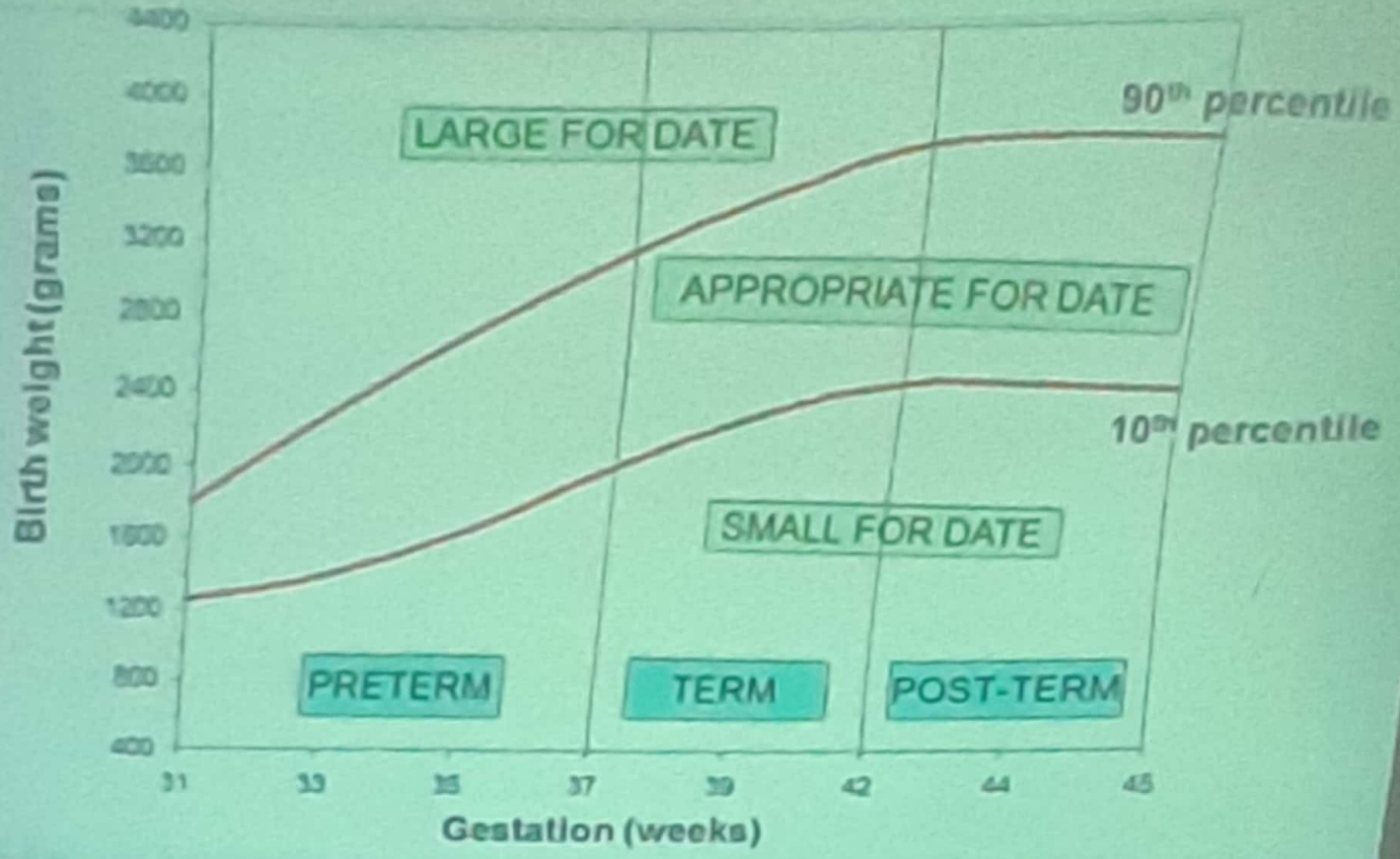
Intrauterine growth chart



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Identification: SFD / IUGR

2.1 Kg - IUGR

3.2 Kg - AFD

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LBW (Preterm) : Problems

- Birth asphyxia
- Hypothermia
- Feeding difficulties
- Infections
- Hyperbilirubinemia
- Respiratory distress
- Retinopathy of prematurity
- Apneic spells
- Intraventricular hemorrhage
- Hypoglycemia
- Metabolic acidosis

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LBW (SFD) : Problems

- Birth asphyxia
- Meconium aspiration syndrome
- Hypothermia
- Hypoglycemia
- Infections
- Polycythemia

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LBW: Issues in delivery

- Transfer mother to a well-equipped centre before delivery
- Skilled person needed for effective resuscitation
- Prevention of hypothermia - topmost priority

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LBW: Indications for hospitalization

- Birth weight <1800 g
- Gestation <34 wks
- Unable to feed*
- Sick neonate*

* Irrespective of birth weight and gestation

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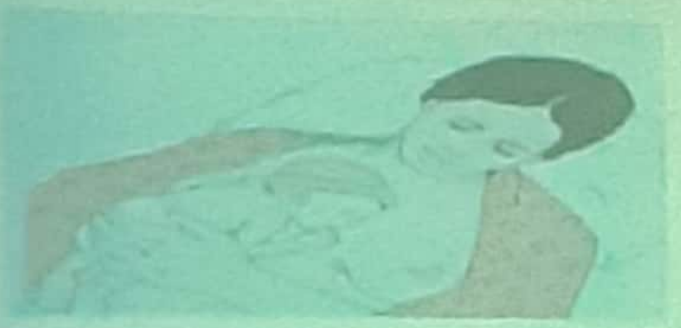
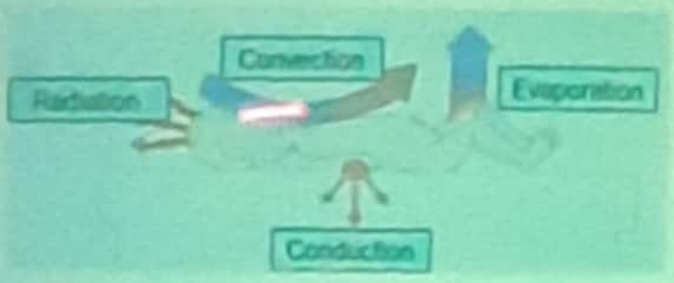
LBW: Keeping warm



Skin-to-skin contact

Birth weight (Kg)	Room temperature (°C)
1.0 – 1.5	34 – 35
1.5 – 2.0	32 – 34
2.0 – 2.5	30 – 32
> 2.5	28 - 30

Warm room, fire or heater



Prevent heat losses

Baby warmly wrapped

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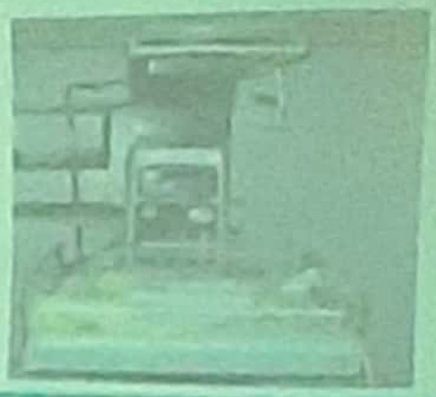
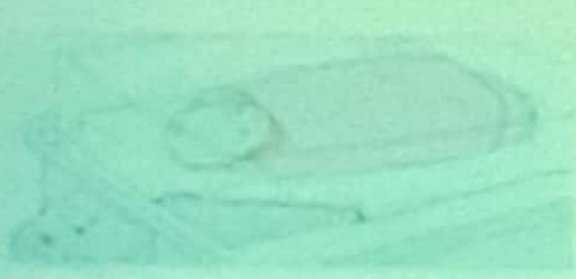
LBW: Keeping warm in hospital

- Skin-to skin method
- Warm room, fire or electric heater
- Warmly wrapped



Radiant warmer



Heated water-filled mattress

Air-heated incubator

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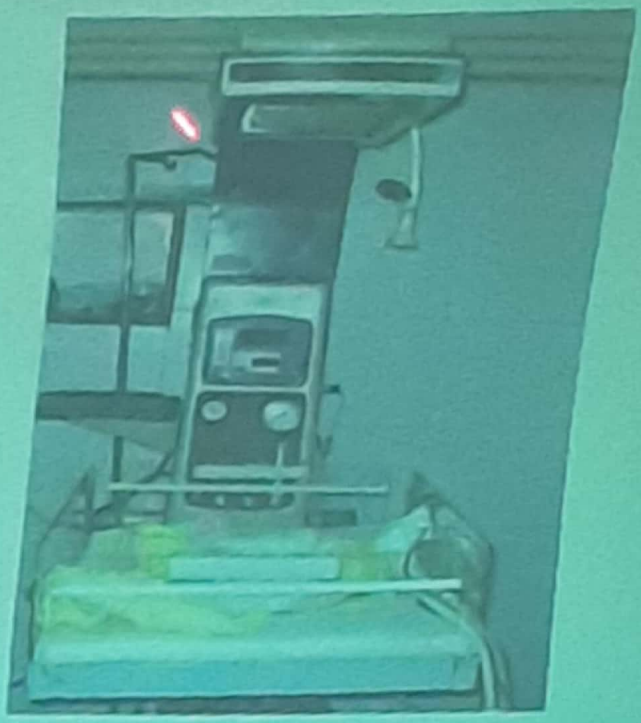
LBW: Keeping warm in hospital

Overhead
Radiant warmer



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LBW: Fluids and feeding

Weight <1200 g; Gestation <30 wks*

- Start initial intravenous fluids
- Introduce gavage feeds once stable
- Shift to spoon feeds over next few days. Later on breast feeds

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* May try gavage feeds, if not sick

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LBW: Fluids and feeding

Weight 1200-1800 g; Gestation 30-34 wks*

- Start initial gavage feeds
- NGT/ Spoon feeding after 1-3 days
- Shift to breast feeds as soon as baby is able to suck

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* May need intravenous fluids, if sick

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LBW: Fluids and feeding

Weight >1800 g; Gestation > 34 wks*

- Breast feeding
- NGT/Spoon feeding, if sucking not satisfactory on breast
- Shift to breast feeds as soon as possible

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LBW: Feeding

Gavage feeding

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Fluid requirements (ml/kg)

Day of life	Birth Weight	
	>1500 g	1000 – 1500g
1	60	80
2	75	95
3	90	110
4	105	125
5	120	140
6	135	155
7 onwards	150	170

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LBW: Adequacy of nutrition

Weight pattern*

- Loses 1 to 2% weight every day initially
- Cumulative weight loss 10%; more in preterm
- Regains birth weight by 10-14 days

Excessive loss or inadequate weight

- Cold stress, anemia, poor intake, sepsis

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LBW: Supplements

- Vitamins : IM Vit K 1.0 mg at birth
  Vit A* 1000 I.U. per day
  Vit D* 400 I.U. per day

- Iron : Oral 2 mg/kg per day from 8 weeks of age

*From 2 weeks of age

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Danger signals (Early detection and referral)

- Lethargy, refusal to feed
- Hypothermia
- Tachypnea, grunt, gasping, apnea
- Seizures, vacant stare
- Abdominal distension
- Bleeding, icterus over palms/soles

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Prognosis

- Mortality
  - Inversely related to birth weight and gestation
  - Directly related to severity of complications
- Long term
  - Depends on birth weight, gestation and severity of complications

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